17 February 1984

MILITARY SPECIFICATION
SURFACE ACOUSTIC WAVE DEVICES, BANDPASS FILTER

This specification is approved for use by US Army Electronics Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the SAWD also described herein shall consist of this document and the latest issue of specification MIL-S-49433(ER).

1. SCOPE

1.1 Scope. This specification covers the detail requirements of a surface acoustic wave (SAW) bandpass filter. The device is a SAW delay line having a center frequency of 96 MHz and a -3 dB bandwidth of 0.47 MHz.

1.2 Part Number. The part number shall be M49433/001FL plus the product assurance level B or S.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solication form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-C-39012 - Connector, Coaxial, Radio Frequency, General Specification for.

MIL-A-46146 - Adhesive-Sealants, Silcone, RTV, Non-corrosive (For Use With Sensitive Metals and Equipments).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Electronics Research and Development Command, ATTN: DELET-R-S, Fort Monmouth, NJ 07703 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 5999

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MIL-S-49433(ER) - Surface Acoustic Wave Devices, General Specifications for.

STANDARDS

MILITARY

MIL-STD-202 - Test Methods for Electronic and Electrical Equipment.

MIL-STD-883 - Test Methods and Procedures for Microelectronics.

3. REQUIREMENTS

3.1 Detail specification. The requirements of the SAW device shall be in accordance with MIL-S-49433(ER) and as specified herein (see figure 1).

3.2 Design, construction and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-S-49433(ER) and as specified herein (see figure 1).

3.2.1 Material.

3.2.1.1 Crystal. The crystal material shall be ST quartz.

3.2.1.2 Adhesive. The adhesive used in bonding the crystal substrate to the chassis shall be in accordance with MIL-A-46146.

3.2.1.3 Solder. Solder shall be in accordance with QQS-571(ER).

3.2.1.4 Internal hook up wire. Internal hook up wire shall be in accordance with MIL-S-49433(ER).

3.2.1.5 Package and sealing material. Package and sealing material shall be in accordance with MIL-S-49433(ER).

3.2.1.6 Connectors (external). Connectors used to effect external connections shall be series SMA and shall conform to MIL-C-39012.

3.3 Electrical requirements. The SAW device shall meet the following electrical requirements:

3.3.1 Return loss. The return loss shall be $\geq$ 9 dB for 86 $\leq$ f $\leq$ 106 MHz when measured as specified in 4.4.1.

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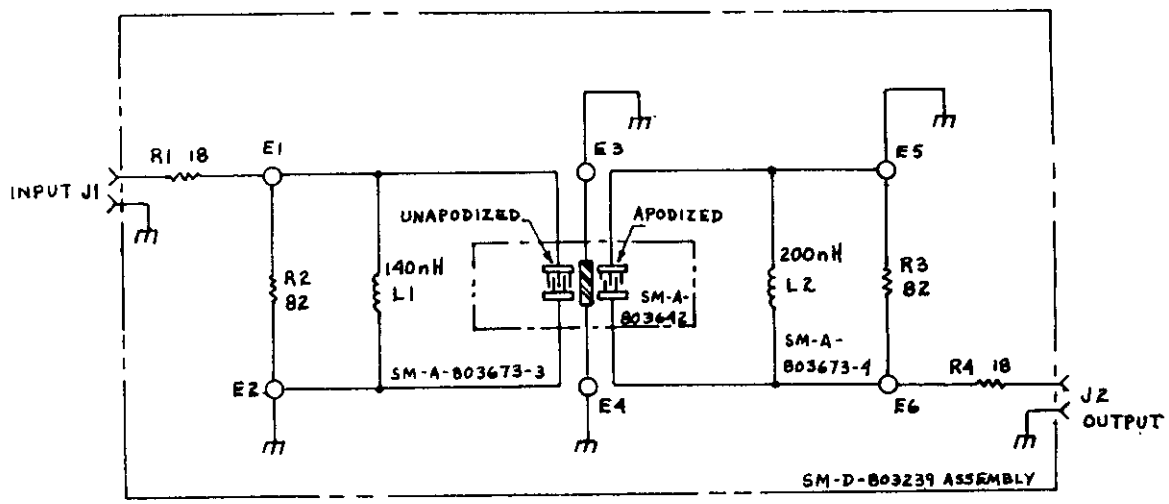


Figure 1. Bandpass filter schematic [1]

[1] See Engineering Drawing for detail

3.3.2 Insertion loss. When measured as specified in 4.4.2 the insertion loss shall be 32 dB ± 1 dB.

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3.3.3 Frequency response width. When measured as specified in 4.4.3 the frequency response width shall be 0.47 MHz ± 0.05 MHz.

3.3.4 Spurious signal. When measured as specified in 4.4.4, the maximum spurious signal shall be ≥ 45 dB down from peak response for 86 ≤ f ≤ 106 MHz.

3.3.5 Group delay. When measured as specified in 4.4.5, the group delay shall be 3.18 s ± 0.05 s.

3.3.6 Time spurious signal. When measured as specified in 4.4.6 the time spurious signal shall be as indicated below:

$To \leq t \leq To + 1.5$ s	≤ -55 dBC
$T_o + 7$ s ≤ t ≤ $T_o + 12.5$ s	≤ -50 dBc
$t \geq T_o + 12$ s	≤ -65 dBc

where T_o is the leading edge of a 1 _ 0.01 s input pulse (see figure 2).

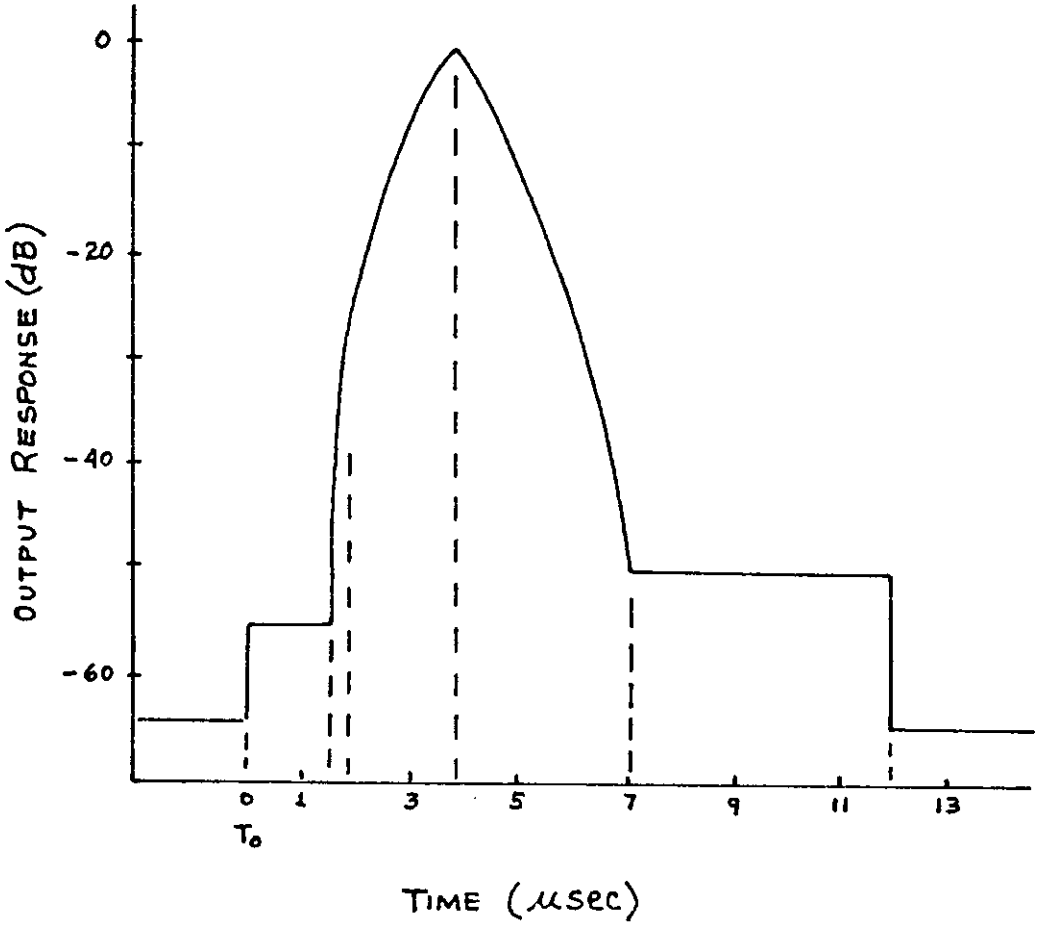


Figure 2. Output response versus time.

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3.3.7 Voltage standing wave ratio (VSWR). Not applicable.

3.4 Environmental requirements. The SAW devices shall meet the environemental requirements of MIL-S-49433(ER).

3.4.1 Operating temperature. The device shall be capable of operating at -55°C to +85°C (see 4.5.1).

3.4.2 Non-operating temperature. The device shall be capable of withstanding -65°C to +125°C under non-operating conditions (see 4.5.2).

3.5 Marking. Marking shall be in accordance with MIL-S-49433(ER).

3.5.1 Serialization. All class S devices shall be serialized in accordance with MIL-S-49433(ER).

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection procedures shall be in accordance with MIL-S-49433(ER).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-S-49433(ER).

4.3 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-S-49433(ER).

4.4 Methods of inspection.

4.4.1 Return loss. The return loss shall be measured by adjusting the matching coils to center tune the input and output transducers. Measure the return loss of both ports of the delay line. The SAW device shall meet the requirements of 3.3.1.

4.4.2 Insertion loss. The insertion loss shall be measured at 96 MHz to meet the requirements of 3.3.2.

4.4.3 Frequency response width. The -3 dB width of the frequency response shall be measured to verify the requirement in 3.3.3.

4.4.4 Spurious signal. The out of band spurious signal shall be measured to verify the requirements of 3.3.4.

4.4.5 Group delay. Group delay shall be measured over a 0.5 MHz bandwidth centered at 96 MHz to meet the requirements of 3.3.5.

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4.4.6 Time spurious signal. The time spurious signal shall be measured to verify the requirements of 3.3.6.

4.5 Temperature.

4.5.1 Operating temperature. The SAW device shall be exposed to a temperature of -55° ± $^{0}_{3}$ °C for a minimum of one hour. At the end of this exposure period and while the SAW device is at this temperature, the insertion loss shall be measured as specified in 4.4.2. The SAW device shall then be exposed to +85° ± $^{3}_{0}$ °C for a minimum of one hour. At the end of this exposure period, and while the SAW device is at this temperature, the insertion loss shall be measured as specified in 4.4.2.

4.5.2 Non-operating temperature. The SAW device shall be exposed to -65° ± $^{0}_{3}$ °C for a minimum of 4 hours. The temperature shall then be raised to 125° ± $^{3}_{0}$ °C for a minimum of 4 hours. At the end of this exposure period, the SAW device shall be stabilized at ambient temperature and the insertion loss shall be measured as specified in 4.4.2.

5. PACKAGING.

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-S-49433(ER).

6. NOTES

6.1 Notes. The notes specified in MIL-S-49433(ER) are applicable to this specification.

Custodian
ARMY - ER

Preparing Activity
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(Project 5999-A079)

Review Activity
Army - AV

User Activity
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